

SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 04.1

Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB.

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Aim

To implement and analyze the experiment described in the official ECA09 MATLAB lab manual.

Procedure / Calculations

Follow the experiment workflow given in the official lab manual: initialize MATLAB, define the specified signal/filter parameters, execute the algorithm, plot the required responses, and record the result.

MATLAB Program

```
clc;
clear;
close all;
%% --- Input Parameters ---
fp = 1000;          % Passband cutoff (Hz)
fs = 4000;          % Sampling frequency (Hz)
N = 50;             % Filter order (even number preferred)
wc = fp/(fs/2);     % Normalized cutoff frequency (0 to 1)
%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect');
%% --- Plot Frequency Response ---
figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');
```

Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'low', rectwin(N+1));
disp('--- Rectangular Window Coefficients ---');
disp(b_rect);

%% --- 2. FIR Low Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'low', hamming(N+1));
disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

%% --- 3. FIR Low Pass using Hanning Window ---
b_hann = fir1(N, wc, 'low', hanning(N+1));
disp('--- Hanning Window Coefficients ---');
disp(b_hann);

%% --- Plot Frequency Responses ---

figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');

figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');

figure;
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');

OUTPUT;
```

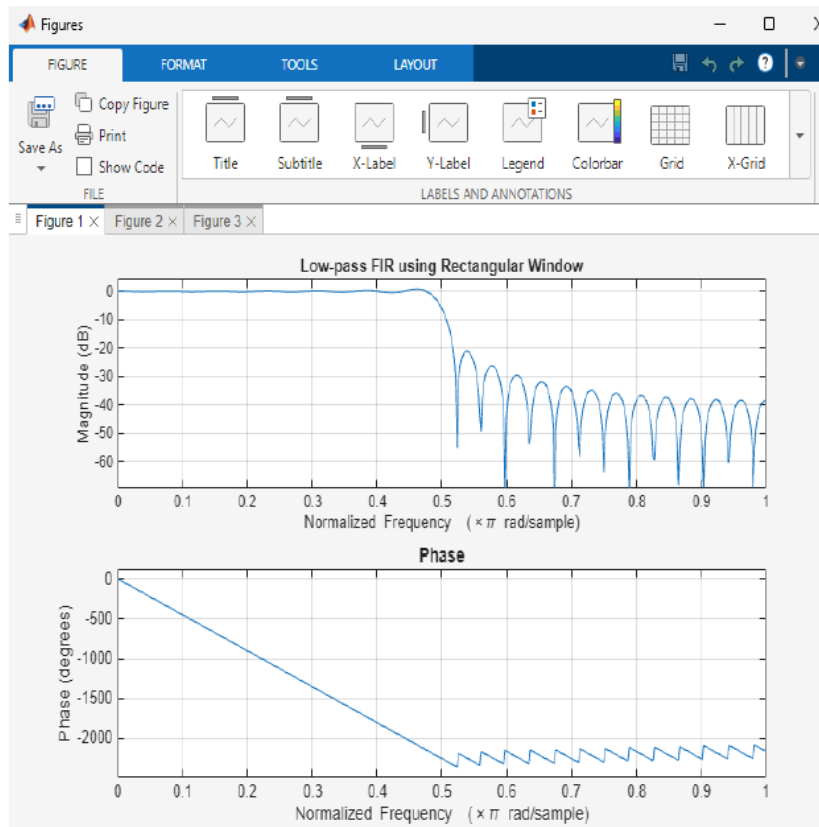
Objective 1:

```
--- Rectangular Window Coefficients ---
0.0126
-0.0000
-0.0137
0.0000
```

0.0150
-0.0000
-0.0166
0.0000
0.0185
-0.0000
-0.0210
0.0000
0.0242
-0.0000
-0.0286
0.0000
0.0349
-0.0000
-0.0449
0.0000
0.0629
-0.0000
-0.1048
0.0000
0.3145
0.4940
0.3145
0.0000
-0.1048
-0.0000
0.0629
0.0000
-0.0449
-0.0000
0.0349
0.0000
-0.0286
-0.0000
0.0242
0.0000
-0.0210
-0.0000
0.0185
0.0000
-0.0166
-0.0000
0.0150
0.0000

-0.0137
-0.0000
0.0126

Objective 2:



Result

1. An FIR Low-Pass Filter was successfully designed using the Rectangular Window technique for the specified cutoff frequency, sampling frequency, and filter order.
2. The FIR filter coefficients were successfully generated using MATLAB and the designed filter was implemented.
3. The magnitude response and phase response of the designed FIR Low-Pass Filter were obtained and analyzed. The magnitude response exhibited low-pass characteristics, while the phase response demonstrated the expected linear-phase behavior.

Practical Question 1

Question:

Design an FIR Low-Pass Filter using the **Rectangular Window** technique in MATLAB for a sampling frequency of **8000 Hz**, cutoff frequency of **1000 Hz**, and filter order of **50**.

Tasks:

1. Calculate the normalized cutoff frequency.
2. Generate the FIR filter coefficients using the `fir1()` function with a Rectangular Window.
3. Plot the magnitude response and phase response of the designed filter.
4. Analyze the low-pass filtering characteristics based on the obtained responses.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Practical Question 2**Question:**

Design an FIR Low-Pass Filter using the **Rectangular Window** technique in MATLAB for a sampling frequency of **12000 Hz**, cutoff frequency of **1800 Hz**, and filter order of **60**.

Tasks:

1. Design the FIR Low-Pass Filter using the specified parameters.
2. Generate and display the FIR filter coefficients.
3. Obtain the magnitude and phase responses using the `freqz()` function.
4. Evaluate the passband and stopband characteristics and compare the response with the ideal low-pass filter behavior.

Aim (10)	Program (20)	Procedure/Calculations (20)	Results (20)	Record (10)	Viva (20)	Total (100)

Result

To design an FIR low-pass filter with the rectangular window, obtain its coefficients, and analyze the magnitude and phase response.

Student Details

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